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Assessing the accuracy of in-stadium and portable multi-camera markerless motion capture for baseball pitching kinematics and kinetics

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ABSTRACT

Markerless (ML) motion capture has emerged as a viable option to marker-based (MB) motion capture in estimating movement biomechanics, but limited data exists on the accuracy of ML systems during high-speed throwing. This study evaluated the accuracy and reliability of an in-stadium (Hawk-Eye) and a portable (Theia3D) ML motion-capture system in quantifying baseball pitching kinematics and kinetics relative to an MB reference. Eighteen collegiate pitchers were simultaneously recorded using all three systems. Mean per-joint position error (MPJPE), statistical parametric mapping (SPM), root mean square error (RMSE), Bland-Altman analysis, and concordance correlation coefficients (CCC) were used to assess agreement. Both ML systems demonstrated measurable discrepancies across variables, with MPJPE values of 56.6 ± 9.4 mm (Hawk-Eye) and 52.0 ± 12.3 mm (Theia3D). Stride length exhibited the strongest agreement with MB in both systems ($CCC > 0.85$), whereas shoulder rotational variables showed greater variability. Error magnitudes in joint positions and kinematic waveforms were comparable to those reported for other ML systems during dynamic movements. These results highlight the influence of system configuration, camera deployment, and pose-estimation models on biomechanical accuracy. Overall, both configurations showed potential for estimating pitching biomechanics, underscoring the trade-offs between criterion and ecological validity in markerless motion capture.

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KEYWORDS

Hawk-Eye; Theia3D; criterion validity; Bland-Altman analysis; statistical parametric mapping

Introduction

Baseball pitching is a form of overarm throwing that involves a complex kinetic chain through which mechanical energy flows to accelerate the throwing arm and ball (Aguinaldo & Escamilla, 2022; Alderink et al., 2021). Motion capture technologies are instrumental in measuring the 3D kinematics and kinetics of joint motion to aid players, coaches, researchers and clinicians in evaluating pitching performance as well as the risk of pitching-related joint injuries (Bullock et al., 2021; Seroyer et al., 2010). Historically, marker-based (MB) motion capture systems have been employed to reliably capture human movement tasks such as baseball pitching. However, the emergence of novel markerless (ML) motion capture systems has allowed for the estimation of 3D segmental pose of human movement without the encumbrance of reflective markers or controlled laboratory settings. Machine-vision based technologies facilitate the collection of kinematic data across more natural settings in which movements such as walking, running, cutting, and jumping can be unobtrusively captured

(Carvalho et al., 2024; Kanko, Laende, Davis, et al., 2021; Mauntel et al., 2021; Ripic et al., 2023; Song et al., 2023).

Recently, in-stadium ML motion capture technology has emerged as the primary modality for ball and bat tracking, video replay, player tracking, and human pose estimation of athletic movements at sporting events such as tennis and baseball (Fitzpatrick et al., 2024; Thomas et al., 2017). In 2020, Major League Baseball (MLB) updated its Statcast (MLB Advanced Media) tracking technology with the installation of Hawk-Eye (Sony Corp., Hawk-Eye Innovations Ltd., Basingstoke, UK) cameras in all MLB stadiums. Each ballpark is equipped with 12 Hawk-Eye cameras, 5 of which are high-frame-rate cameras dedicated to pitch and bat tracking, operating at 300 Hz (MLB, 2024). The system allows for in-game biomechanical tracking of players using skeletal poses estimated by Hawk-Eye's machine learning models driven by images from these high-speed cameras.

Other multi-camera ML motion capture systems similarly track the pitch, implement, and player biomechanics, including KinaTrax (Sony Corp., KinaTrax, Inc., Boca

Raton, FL), Theia3D (Theia Markerless, Inc., Kingston, Ontario, Canada), SIMI (Simi Reality Motion Systems, Unterschleißheim, Germany), and DARI Motion (DARI Motion, Overland Park, KS). However, the criterion validity and reliability of these systems reported in the relevant literature have been limited to simple motions such as walking (Kanko, Laende, Strutzenberger, et al., 2021; Ripic et al., 2022), running (Song et al., 2023; Tang et al., 2024), and jumping (Mauntel et al., 2021; Templin et al., 2024). Furthermore, while research on the application of ML motion capture technology in providing in-game pitching insights has emerged (Giordano et al., 2024; Lerch et al., 2024), concurrent assessments of agreement in pitching kinematics between ML and MB motion capture technologies have been limited so far to DARI Motion (Fleisig et al., 2022) and pitchAI (3MotionAI, Inc., Oakville, ON, Canada), which is a monocular ML solution designed specifically to capture the pitching motion with a smartphone (Dobos et al., 2022). There are no known studies that have reported the performance of an in-stadium multi-camera ML system in skeletal pose estimation. Moreover, the validity of pitching-related kinetics from any ML system remains unexamined.

Therefore, the aim of this study was to evaluate the accuracy of an in-stadium Hawk-Eye ML motion capture system and a portable, lab-based Theia3D ML system in measuring baseball pitching kinematics and kinetics using traditional MB motion capture as the reference standard. Operational constraints of in-stadium systems such as Hawk-Eye including fixed, distant camera placement and a sparse keypoint model (Michaelides, 2025) would reasonably be expected to reduce skeletal pose estimation accuracy compared to portable systems like Theia3D, which allow flexible camera placement and a denser keypoint configuration (Kanko, Laende, Davis, et al., 2021). However, the extent and practical implications of these contrasting system features, particularly during high-speed movements such as baseball pitching, have not been empirically tested.

Methods

Participants

Using the method proposed by Lu et al. (2016) for Bland-Altman analysis, a sample size of 16 was estimated based on a confidence interval (CI) half-width of 5° around the limits of agreement (LoA) and a standard deviation of 7° for joint angle differences as reported in a prior validation of another ML system (Fleisig et al., 2022). To account for data dropout, 18 collegiate male baseball players (height = 1.83 ± 0.02 m, mass = 86 ± 18 kg) from

NCAA Division I and Division II programmes provided written informed consent to participate in this study, which was approved by the university's Institutional Review Board (IRB# 18421). After each pitcher completed a self-supervised warm-up, 38 reflective markers (9 mm DIA) were placed according to a previously described link-segment model (Aguinaldo et al., 2007). A static calibration trial was performed with the pitcher standing on the mound with the shoulders abducted 90° (T-pose) facing home plate.

Data acquisition

All pitchers threw 10 maximum effort fastballs off the mound of an MLB baseball stadium while markerless (ML) and marker-based (MB) motion capture data were concurrently captured using a 5-camera in-stadium ML motion capture system (Hawk-Eye Innovations Ltd., Basingstoke, UK) and an 18-camera hybrid (ML and MB) motion capture system (Qualisys, Göteborg, Sweden). The five high-speed Hawk-Eye cameras were permanently mounted around the playing field with pairs of cameras positioned in the stands on each of the first and third base sides as well as one above the centre field wall with their fields of view facing the pitching mound (Figure 1a). The Qualisys system contained a network of 10 ML high-speed cameras (Miquis, Qualisys, Göteborg, Sweden) genlocked with 8 MB visible-red cameras (Arqus, Qualisys, Göteborg, Sweden) and were positioned on tripods surrounding the pitching mound within a 4-meter radius (Figure 1b). The Hawk-Eye cameras were calibrated to a global reference frame with its origin at home plate while the Qualisys cameras were calibrated and synchronised to a separate global reference frame, independent of Hawk-Eye, with its origin located on the pitching rubber.

During data collection, the in-stadium ML system was operated by trained Hawk-Eye technicians while the research team operated the hybrid ML and MB Qualisys system. Although the Hawk-Eye and Qualisys systems were independently triggered, motion data were captured at 300 Hz at full resolution concurrently by cameras in both systems. We used the pitch numbers and timestamps as well as the release speeds and spin rates from a radar-based ball tracking unit (TrackMan B1, TrackMan, Hoersholm, Denmark) as metadata to match all pitching captures from both systems.

Markerless motion capture

Video images from the in-stadium cameras were processed in Hawk-Eye's machine-vision software (v. 2024.1.26.3) and 3D coordinate data for 29 keypoints

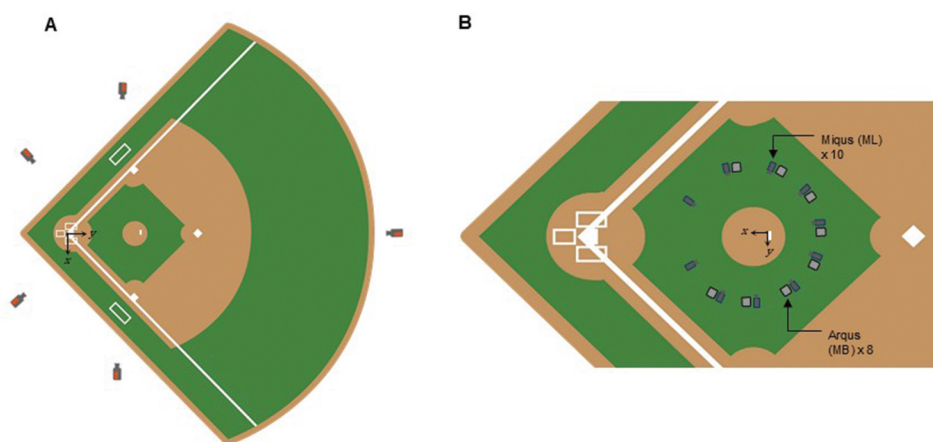


Figure 1. Camera configurations for the A) in-stadium (Hawk-Eye) and B) portable hybrid (MB+ML and Theia3D) motion capture systems.

were outputted for each pitch. Hawk-Eye employs a machine learning model optimised with inverse kinematics (IK) to estimate segmental poses to which the bilateral locations of the ankle, knee, hip, shoulder, elbow, and wrist joint centres as well as other keypoints on the body can be located throughout the pitch (Figure 2a). A customised pipeline in Visual3D (HAS-Motion, Canada) was used to transform the 29 keypoints from Hawk-Eye's global coordinate system to the Qualisys reference frame using the mid-hip point as the root to spatially align the data from the two systems. Similarly, we processed the 10 synchronised videos from the Miquis cameras in Theia3D (v. 2023.1.0.3160, Theia Markerless, Kingston, ON) to perform 3D pose estimation. The commercially available software uses a deep learning approach to approximate 124 salient features on the body and a scaled IK model (Figure 2b) to solve the segment positions of each pitcher (Kanko, Laende,

Davis, et al., 2021, 2021; Song et al., 2023). The pose estimation data were outputted as 4×4 matrices and merged with the tracked marker trajectories from the marker-based motion capture using Visual3D.

Data analysis

All positional data were digitally processed using a fourth-order, zero-lag, double-pass Butterworth filter at a cut-off frequency of 18 Hz (Zheng et al., 2004). Because the Hawk-Eye, Theia3D, and MB systems each use different sets of keypoints or markers, we replicated the Theia3D IK model, which included the feet, shanks, thighs, pelvis, trunk, head, upper arms, forearms, and hands, and scaled it to align with the joint centres and keypoints available in each system's dataset. The kinematic chain was constrained to have six degrees-of-freedom (DOF) at the pelvis and three DOF at the hip,



Figure 2. Example markerless models with A) Hawk-Eye keypoints and B) Theia3D skeleton overlay on original video of a representative pitch.

shoulder, and lumbosacral joints. We modelled the elbow, wrist, and ankle joints as two DOF universal joints and the knee joint as a 1-DOF hinge joint. The throwing hand and baseball (142 g) were modelled together as a rigid segment.

Kinematic measures included the locations of the hip, knee, ankle, shoulder, elbow, and wrist joint centres as well as joint angles and rotational velocities of the lead knee, pelvis, trunk, throwing shoulder, and throwing elbow. Eight kinematic time series, including lead knee flexion, pelvis rotation, trunk rotation, shoulder internal-external rotation, elbow flexion, pelvis rotational velocity, trunk rotational velocity, and shoulder rotational velocity were estimated from the joint and segment pose data from each motion capture system using previously described definitions (Aguinaldo et al., 2007; Wu et al., 2005). Joint kinetics of the throwing shoulder and elbow were estimated using the same inverse dynamics approach (Aguinaldo et al., 2007; Robertson et al., 2014) with each motion capture system.

Twelve discrete kinematic and kinetic metrics considered to be key performance and injury risk indicators of pitching (Dobos et al., 2022; Fleisig et al., 2022) were extracted for each pitch using Hawk-Eye, Theia3D, and MB data. Three of the 12 parameters (lead knee flexion, trunk rotation, elbow flexion) occurred at the instant of stride (lead) foot contact (SFC), defined as the first frame after the velocity of the stride foot's centre-of-mass descends below 1.0 m/s. The remaining 9 parameters (pelvis rotation, shoulder external rotation, pelvis rotational velocity, trunk rotational velocity, shoulder internal rotational velocity, shoulder distraction force, shoulder external rotation torque, and elbow valgus torque) were their maximum values that occurred in the time between SFC and ball release (BR).

The concurrent validities of the joint centre position data (hip, knee, ankle, shoulder, elbow, wrist) from both ML systems were summarised as pitch-average differences between the ground truth (MB) and predicted (ML) positions as well as average 3D root-mean-square (RMS) differences across all pitches (Kanko, Laende, Davis, et al., 2021). In addition, we employed the mean per-joint position error (MPJPE), which is an established metric used to evaluate the performance of machine-vision pose estimation methods (Gholami et al., 2022; Sarandi et al., 2023). It computes the mean Euclidean distance between the MB and ML derived 3D joint centre positions after root alignment:

$$MPJPE = \frac{1}{n} \sum_{i=1}^n ||y_i - \hat{y}_i|| \quad (1)$$

where n is the number of joints, y_i is the ground truth position of the i -th joint, $\hat{y}_i$ is the predicted position of the i -th joint, and $||\cdot||$ denotes the Euclidean distance.

Kinematic-based events were used to time-normalise the pitching motion, defined from the maximum height of the lead knee (MKH) to follow-through (FT), defined as 100 ms after BR. We performed statistical parametric mapping (SPM{t}) to evaluate differences in the ensemble-averaged kinematic waveforms between each markerless and marker-based system using the *spm1d* (0.4.32) python package (Pataky et al., 2013). Kinematic waveforms from both ML modalities were each compared against the MB data for each pitch using the root-mean-square error (RMSE), which quantifies the average difference between ML and MB modalities over each frame of the time-series data. An ML-derived kinematic signal was considered to demonstrate strong validity if $RMSE \leq 5^\circ$ for joint angles (Lima et al., 2024; Song et al., 2023), $RMSE \leq 20^\circ/s$ for pelvis and trunk rotational velocities (Fleisig et al., 2024), and $RMSE \leq 300^\circ/s$ for shoulder internal rotational velocity (García-Alsina et al., 2005).

To assess the concurrent validity and reliability of the twelve discrete metrics estimated by Hawk-Eye and Theia3D, a nested Bland-Altman analysis with the concordance correlation coefficient (CCC) was implemented using the marker-based motion capture as the criterion method. The analysis produces upper and lower limits of agreement (LoA) for data where there are multiple observations per subject by estimating the 95% confidence intervals (CI) around the bias and each LoA using the method of variance estimates recovery (MOVER) (Zou, 2013). A key feature of this agreement analysis is its capacity to measure both the mean bias and the correlation of two methods using the CCC, which is expressed in terms of the variance components of a linear mixed model (Carrasco et al., 2013; Das et al., 2023). CCC values range from -1 to 1 , with higher values indicating stronger agreement. All data agreement analyses were performed with the *simplyagree* (0.0.3) package in R (Caldwell, 2022).

Results

The raw dataset was reduced to a sample of 104 matching pitches after excluding 69 pitches due to poor marker tracking (MB) and 7 pitches due to image corruption (Theia3D). Although no pitches were excluded due to incomplete Hawk-Eye data, only the subset with corresponding MB and Theia3D data could be retained, ensuring that comparisons across systems were based solely on pitches with complete data from all modalities. Ultimately, data from all 18 pitchers were included, with an average of 7 fastball pitches per pitcher (range 5 to 10). The average fastball speed across all pitches was 135.7 ± 6.6 km/h (84.3 ± 4.1 mph) and upon visual inspection, all pitchers threw with a '3/4' or over-the-top arm slot position.

When benchmarked against MB, both ML systems demonstrated measurable joint position errors that were smaller in the lower extremity (Figure 3) than in the upper extremity (Figure 4). The smallest average errors were observed at the hip (~40 to 43 mm) and the largest at the elbow (70 to 75 mm). Mean per-joint position error (MPJPE) across all joints averaged 56.6 ± 9.4 mm for the in-stadium system and 52.0 ± 12.3 mm for the portable system.

Kinematic waveforms

Ensemble-averaged joint angles and rotational velocities measured with both ML systems are shown against the corresponding MB waveforms in Figure 5a, with the distributions of within-pitch RMS errors displayed in Figure 5b. Across most variables, both systems demonstrated measurable discrepancies relative to MB. RMS errors for joint angles ranged from approximately 2° to 16° , and for rotational velocities from about $50^\circ/\text{s}$ to over $400^\circ/\text{s}$. Theia3D exhibited smaller RMS errors in knee flexion, pelvis rotation, trunk rotation, pelvis rotational velocity, and shoulder rotational velocity. Hawk-Eye showed comparable error magnitudes in elbow flexion and shoulder internal-external rotation and exhibited lower mean error in trunk rotational velocity. For both systems, ML-MB differences in elbow flexion were evident during the stride and follow-through phases (SPM {t} $p < .05$). Shoulder internal – external rotational velocity displayed the largest offsets relative to MB for both systems, particularly for Hawk-Eye.

Discrete kinematic and kinetic metrics

Table 1 summarises the mean $\pm$ SD, bias, limits of agreement (LoA), concordance correlation coefficients (CCC), and 95% confidence intervals (CI) for all discrete metrics, with corresponding Bland – Altman plots shown in Figures 6 and 7. Both markerless systems tended to underestimate stride length with a bias of 0.04 m to 0.05 m relative to MB with strong concordance ($\text{CCC} > 0.85$). Across most kinematic metrics, LoA and CCC values indicated variable agreement with MB, ranging from modest to strong. For joint angles, biases ranged from -2° to 11° for the in-stadium system and -2° to 22° for the portable system, with CCC values spanning -0.07 to 0.93 . LoA for joint angles extended roughly 9° to 50° , and for rotational velocities from roughly $220^\circ/\text{s}$ to $2850^\circ/\text{s}$, reflecting greater variability in high-speed motions. Maximum shoulder internal rotation velocity showed the weakest agreement overall across systems (bias = $573^\circ/\text{s}$, $\text{CCC} = 0.12$).

For discrete kinetic metrics (Figure 7), both systems underestimated maximum shoulder distraction force by as much as 38 N and showed moderate concordance ($\text{CCC} = 0.61$). Estimates of shoulder external rotation torque and elbow valgus torque exhibited larger discrepancies from MB for both systems, with biases ranging from -19 Nm to 39 Nm and CCC values below 0.4, indicating weaker agreement in these kinetic variables.

Discussion

This study evaluated the validity and reliability of baseball pitching kinematics and kinetics measured by two different multi-camera markerless (ML) motion capture systems, Hawk-Eye (in-stadium) and Theia3D (portable), against ground truth data from a marker-based (MB) system. Both systems demonstrated measurable discrepancies with error magnitudes varying across joint positions, kinematic waveforms, and discrete metrics. The overall pattern of results indicated that the portable system produced smaller average errors in many kinematic and kinetic measures, while the in-stadium system performed comparably in several key metrics, including stride length, pelvis rotation, elbow flexion, shoulder internal-external rotation, and trunk rotational velocity. These findings illustrate how differences in system architecture, camera configuration, and keypoint detection modelling may influence accuracy, particularly in high-speed, multi-planar movements such as baseball pitching. Importantly, these results should not be interpreted as demonstrating inherent superiority of one ML system over another, but rather as evidence of how specific design characteristics manifest in measured biomechanical outcomes of baseball pitching. Thus, our findings reflect the distinct operational designs of each ML system rather than a hierarchy of performance accuracy.

While the validity of Theia3D has been evaluated during slower speed movements such as walking, running, and jumping (Kanko, Laende, Davis, et al., 2021; Song et al., 2023; Wren et al., 2023), only DARI Motion (Fleisig et al., 2022) and pitchAI (Dobos et al., 2022) have been validated during baseball pitching. Despite the widespread use of Hawk-Eye in professional baseball (MLB, 2024), this is the first study to assess its performance in estimating skeletal poses. While Hawk-Eye generally produced RMS errors that failed to meet thresholds for strong validity in joint positions and kinematic waveforms, errors in lead knee flexion ($\sim 2^\circ$ to 6°) and elbow flexion ($\sim 13^\circ$) aligned with values reported in other high-speed movements and previous pitching studies (Dobos et al., 2022; Fleisig et al., 2022; Lima et al., 2024; Song et al., 2023). This convergence suggests shared challenges in modelling high-speed movements,

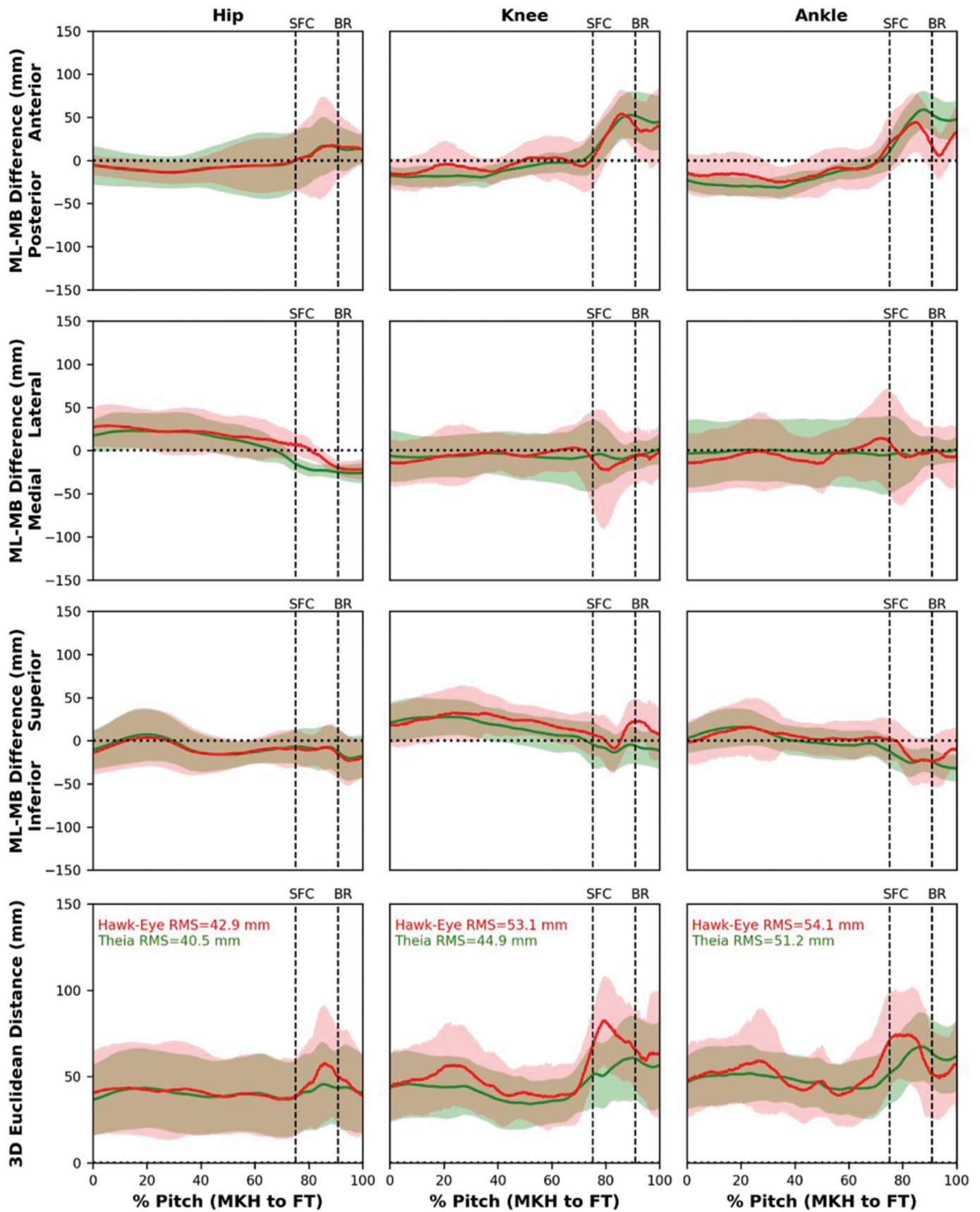


Figure 3. Lower extremity joint position errors (mean $\pm$ SD) for Hawk-Eye (—) and Theia3D (—) relative to the marker-based system across the pitching motion. Errors are shown as differences along each global coordinate axis (rows 1–3) and as 3D Euclidean distances (row 4). RMS errors for each joint are inset in the Euclidean distance plots. MKH = maximum knee height; SFC = stride foot contact; BR = ball release; FT = follow-through.

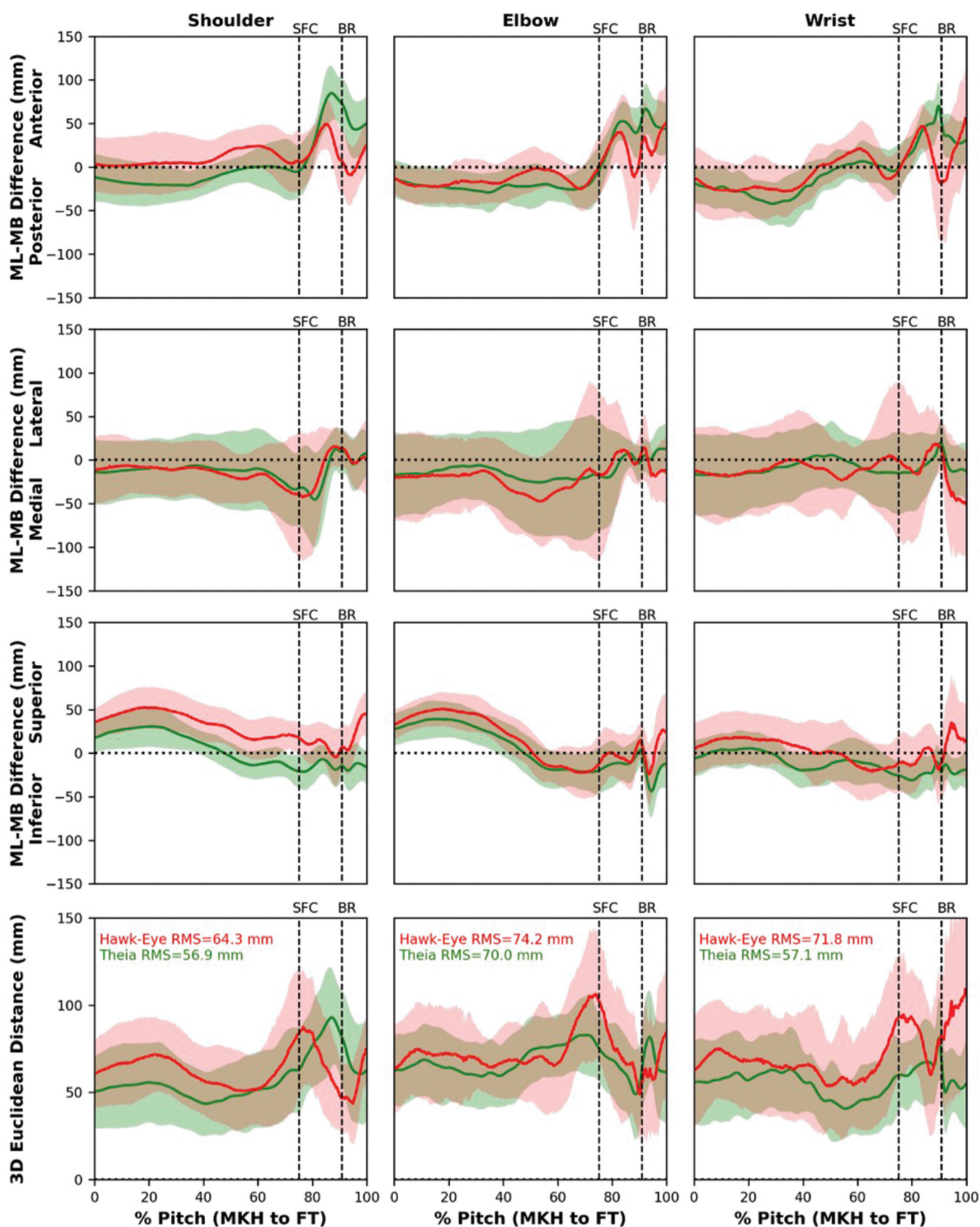


Figure 4. Upper extremity joint position errors (mean $\pm$ SD) for Hawk-Eye (—) and Theia3D (—) relative to the marker-based system across the pitching motion. Errors are shown as differences along each global coordinate axis (rows 1–3) and as 3D Euclidean distances (row 4). RMS errors for each joint are inset in the Euclidean distance plots. MKH = maximum knee height; SFC = stride foot contact; BR = ball release; FT = follow-through.

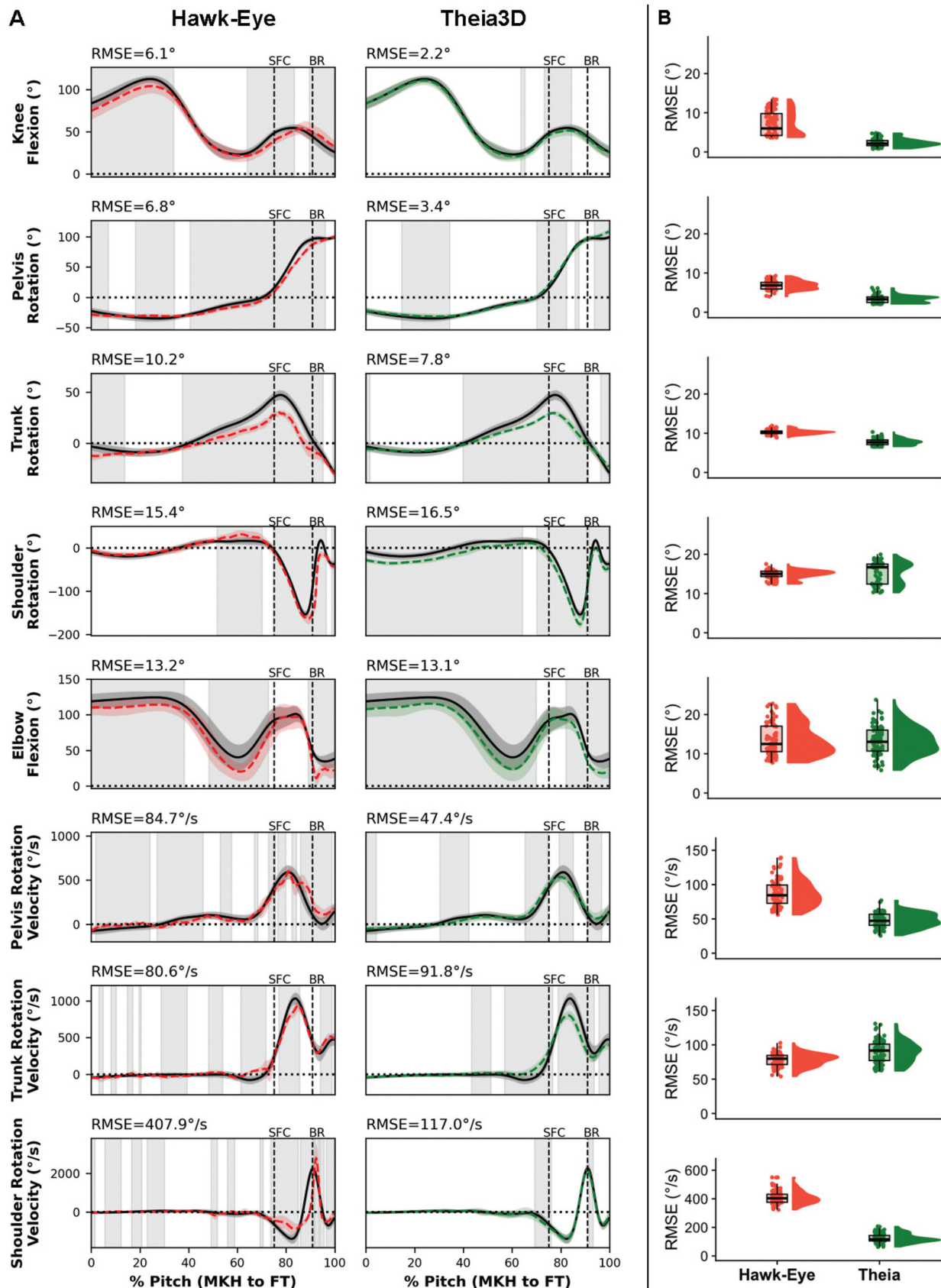


Figure 5. A) ensemble-averaged kinematic plots are shown for Hawk-Eye (—), Theia3D (—), and MB (—) systems with ± 1 SD clouds (shaded areas) and supratherreshold (SPM{t} $p < .05$) clusters (filled bands) across the pitching motion. RMSE values shown in each panel represent the mean of the within-pitch RMS errors computed for each pitch relative to the MB reference, averaged across all pitches. B) corresponding RMSE raincloud plots show the distributions of errors in kinematics derived with Hawk-Eye and Theia3D. MKH = maximum knee height; SFC = stride foot contact; BR = ball release; FT = follow-through.

Table 1. Bland-Altman analysis of discrete kinematic and kinetic metrics. Data include means and standard deviations for marker-based, Hawk-Eye, and Theia3D systems, biases, limits of agreement (LoA), and concordance correlation coefficient (CCC) values for reliability and agreement between marker-based and markerless systems.

Metric	Mean (SD)		Bias (CI)		Lower LoA (CI)		Upper LoA (CI)		CCC (CI)	
	Marker-based		Hawk-Eye	Theia3D	Hawk-Eye	Theia3D	Hawk-Eye	Theia3D	Hawk-Eye	Theia3D
Stride Length (m)	1.4 (0.11)	1.4 (0.11)	1.4 (0.11)	1.4 (0.13)	-0.05 (-0.06, -0.03)	-0.04 (-0.05, -0.02)	-0.09 (-0.12, -0.07)	-0.08 (-0.11, -0.06)	-0.01 (-0.02, 0.03)	0.00 (-0.02, 0.04)
Lead Knee Flexion at SFC (°)	50.9 (9)	45.5 (9.8)	46.9 (8.1)	46.9 (8.1)	-5.2 (-9.9, -0.6)	-3.3 (-7.0, 0.4)	-23.5 (-34.7, -16.9)	-17.8 (-26.8, -12.5)	13.0 (6.4, 24.2)	11.2 (5.9, 20.2)
Trunk Rotation at SFC (°)	49.1 (8)	37.3 (12)	32.3 (7.1)	32.3 (7.1)	-11.3 (-15.8, -6.8)	-16.2 (-18.5, -13.9)	-30.0 (-40.6, -23.8)	-26.0 (-31.4, -22.8)	7.5 (1.2, 18.0)	-6.4 (-9.6, -1.0)
Elbow Flexion at SFC (°)	94.0 (16.5)	85.7 (15.9)	91.6 (16.3)	91.6 (16.3)	-3.6 (-5.8, -1.3)	-2.3 (-3.5, -1.2)	-18.7 (-23.3, -15.7)	-10.7 (-13.1, -9.1)	11.6 (8.6, 16.1)	6.1 (4.5, 8.5)
Maximum Pelvis Rotation (°)	35.3 (4.7)	32.3 (2.3)	31.4 (2.7)	31.4 (2.7)	-2.4 (-3.1, -1.7)	-3.4 (-4.4, -2.4)	-6.9 (-8.3, -5.9)	-8.2 (-10.4, -6.9)	2.0 (1.1, 3.5)	1.4 (0.1, 3.6)
Maximum Shoulder External Rotation (°)	159 (11.4)	168.6 (10.1)	177.6 (6.9)	177.6 (6.9)	11.1 (4.4, 17.9)	21.7 (16.5, 26.8)	-14.0 (-31.1, -4.5)	2.6 (-10.4, 9.8)	36.3 (26.8, 53.3)	40.8 (33.6, 53.7)
Maximum Pelvis Rotation Velo (°/s)	602.7 (81.4)	606.5 (31.5)	546.1 (55.2)	546.1 (55.2)	5.3 (-11.5, 22.2)	-59.2 (-76.0, -42.3)	-162.5 (-203.1, -134.0)	-170.0 (-204.3, -147.6)	173.2 (144.7, 213.8)	51.7 (29.3, 85.9)
Maximum Trunk Rotation Velo (°/s)	1038.4 (77.1)	951 (70.5)	813.2 (49.3)	813.2 (49.3)	-89.0 (-124.1, -53.9)	-232.7 (-245.3, -220.0)	-270.2 (-345.0, -224.6)	-336.1 (-362.7, -317.8)	92.2 (46.6, 167.0)	-129.2 (-147.6, -102.7)
Maximum Shoulder IR Velo (°/s)	2289.1 (168.9)	2818.2 (711.9)	2188.3 (182.1)	2188.3 (182.1)	573.7 (294.0, 853.3)	-118.5 (-180.0, -56.9)	-850.0 (-1448.9, -485.8)	-602.0 (-729.8, -514.2)	1997.4 (1633.2, 2596.2)	365.0 (277.3, 492.9)
Maximum Shoulder Distraction Force (N)	222.1 (102.8)	196.9 (23.5)	201.5 (48.1)	201.5 (48.1)	-37.4 (-51.1, -23.8)	-38.0 (-47.9, -28.0)	-163.1 (-192.4, -142.2)	-116.3 (-136.4, -102.4)	88.2 (67.3, 117.5)	40.4 (26.5, 60.4)
Maximum Shoulder ER Torque (Nm)	95.9 (21.4)	133.8 (15.6)	75.9 (14.3)	75.9 (14.3)	38.8 (30.3, 47.3)	-19.1 (-23.5, -14.6)	-16.9 (-34.0, -5.7)	-53.0 (-62.1, -46.8)	94.5 (83.3, 111.6)	14.9 (8.7, 24.0)
Maximum Elbow Valgus Torque (Nm)	89.4 (7.2)	126 (10.7)	80.6 (13.7)	80.6 (13.7)	33.7 (28.8, 38.7)	-9.8 (-13.0, -6.5)	6.4 (-3.9, 12.8)	-36.5 (-43.6, -31.6)	61.1 (54.6, 71.4)	16.9 (12.1, 24.1)
									-0.00 (-0.09, 0.08)	-0.07 (-0.26, 0.12)

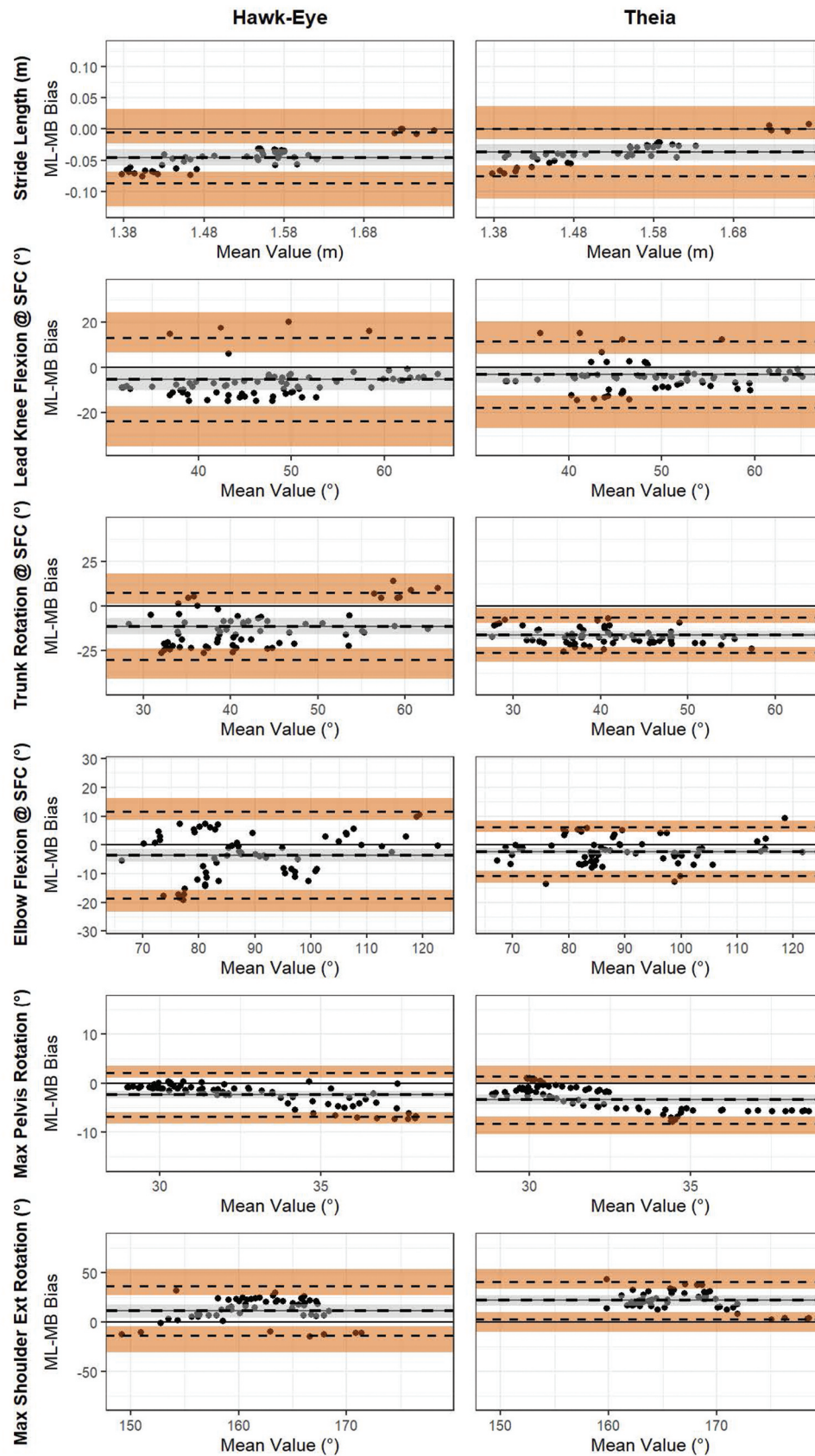


Figure 6. Bland-Altman plots for stride length and joint angle metrics estimated by Hawk-Eye and Theia3D. Shaded areas represent the 95% confidence intervals around the mean bias and limits of agreement.

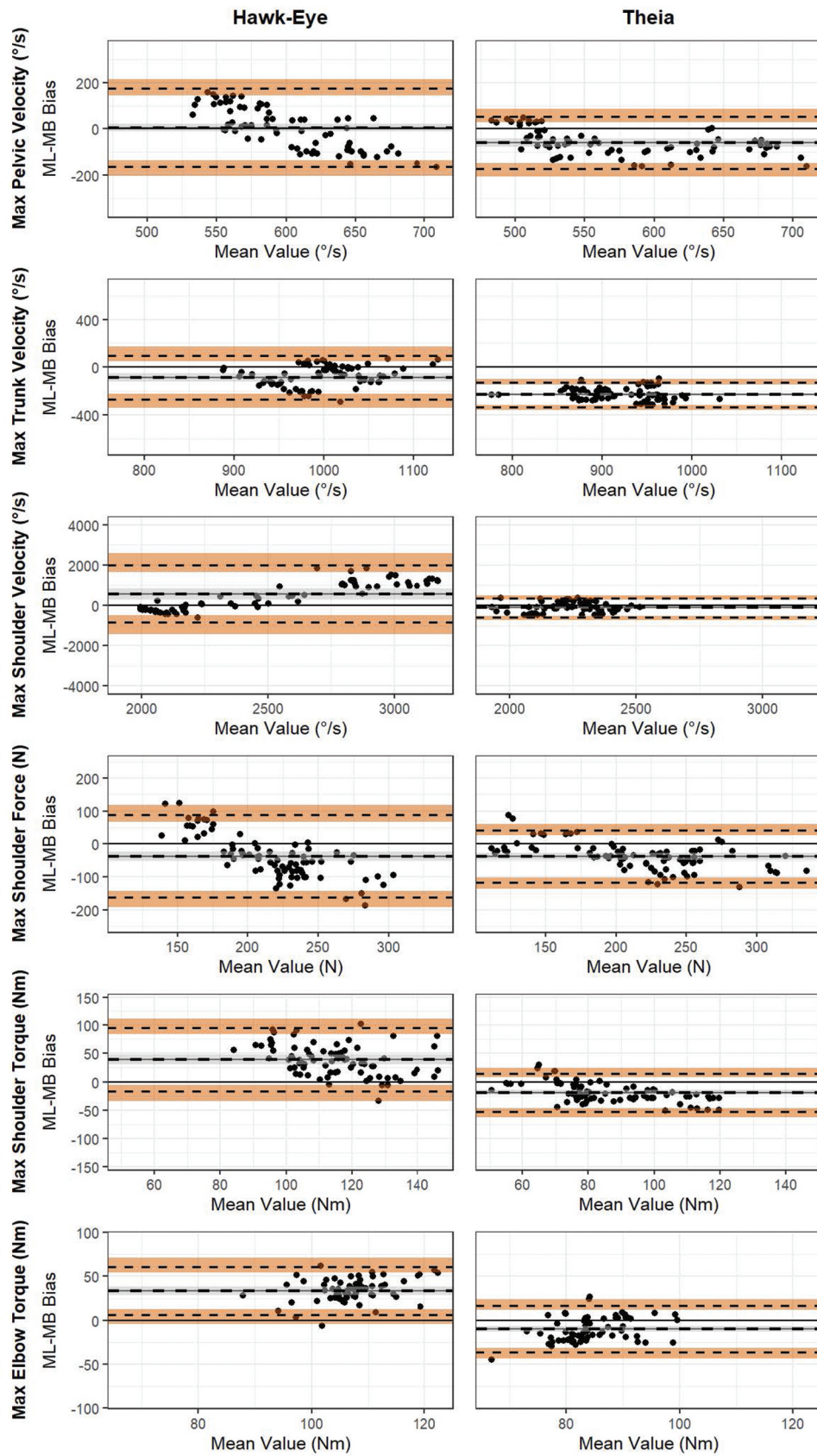


Figure 7. Bland-Altman plots for joint rotational velocity and joint torque metrics estimated by Hawk-Eye and Theia3D. Shaded areas represent the 95% confidence intervals around the mean bias and limits of agreement.

including occlusion, motion blur, and limited training data for overhead throwing (Lerch et al., 2025; Scataglini et al., 2024). Given the limited number of baseball-specific validations, further research is needed to evaluate markerless systems across other tasks such as hitting and fielding, as well as across different system configurations and competitive levels.

Across all transverse plane angles included in the waveform analysis, pelvis rotation exhibited the lowest RMS errors and shoulder internal-external rotation the highest for both ML systems, with values comparable to those previously reported for pitchAI (Dobos et al., 2022). The SPM plots (Figure 5a) also show less waveform agreement between each ML- and MB-derived shoulder internal-external rotations during the arm-cocking and acceleration phases (SFC to BR) of the pitching motion. This level of disagreement in shoulder kinematics may be attributed to differences in upper-extremity joint centre estimation as errors at the shoulder, elbow, and wrist joints were substantially larger than those reported with Theia3D during normal gait (Kanko, Laende, Davis, et al., 2021). These errors, however, align with ranges reported for other markerless systems under more dynamic conditions (Lima et al., 2024; Scataglini et al., 2024) and help explain the downstream variability observed in kinematic and kinetic variables. In particular, the joint position errors likely propagated through inverse kinematics and dynamics, amplifying errors in joint angles, angular velocities, and torques after numerical differentiation despite comparable overall waveform trends. These effects were most evident in rotational variables, especially at the shoulder, where rapid internal rotation magnifies even small spatial deviations (Giordano et al., 2024; Lerch et al., 2024). The larger RMSE in pelvis rotation velocity observed with Hawk-Eye stems from this propagation, as its ensemble-averaged signal exhibited greater frame-to-frame noise, compared to the MB and Theia3D waveforms (Figure 5).

Bland-Altman results demonstrated varying levels of validity and reliability in discrete kinematic (Figure 6) and kinetic (Figure 7) metrics for both markerless systems. Stride length showed the strongest agreement with MB for both, consistent with previous findings with DARI Motion (Fleisig et al., 2022) and pitchAI (Dobos et al., 2022). Mean biases in joint angles (Table 1) were similar in magnitude to those reported by Fleisig et al. (2022), who suggested that systematic offsets could likely be corrected through calibration. More informative were the widths of the limits of agreement (LoA) and their confidence intervals (CI), which reflect the degree to which system estimates deviated from MB across the measurement range. Both systems displayed narrower LoA and higher concordance in

lower-extremity kinematics than in upper-extremity or kinetic variables, where faster motions introduced greater variability. For example, elbow flexion at SFC showed small biases ($< 5^\circ$) and strong concordance ($CCC > 0.8$) in both systems, although the LoA width and CI for Hawk-Eye were nearly twice those of Theia3D (Figure 6), indicating greater dispersion in estimated values. Overall, agreement indicators varied across metrics, reflecting the influence of camera configuration, model architecture, and pose-estimation algorithms on measurement reliability. These findings highlight the importance of continued refinement of ML motion capture systems, particularly in-stadium configurations, for capturing high-speed athletic movements (Lerch et al., 2025; Scataglini et al., 2024).

To our knowledge, this is the first study to evaluate the validity and reliability of joint kinetics derived from any ML system during baseball pitching. Both configurations showed measurable biases and variability in estimating shoulder distraction force, shoulder external rotation torque, and elbow valgus torque relative to MB. Bias magnitudes in kinetic metrics were within the range of those reported for the lower extremity joints with Theia3D during running and cutting (Song et al., 2023). Hawk-Eye exhibited wider limits of agreement and lower CCC values for shoulder and elbow torques, which may reflect a higher sensitivity of kinetic outcomes to upstream kinematic errors, especially in rapidly moving segments of the throwing arm, consistent with previous work showing larger errors in high-speed movements (Kanko, Laende, Davis, et al., 2021; Song et al., 2023).

Several limitations should be considered when interpreting our findings. First, a subset of pitches was excluded due to incomplete marker or image data across the marker-based and portable configurations, which may have introduced selection bias towards more stable tracking conditions. No apparent differences in pitch velocity were observed between retained and excluded trials, though a detailed analysis of exclusion patterns was beyond the present scope. Next, while MB motion capture served as the reference standard, soft-tissue artifacts compromise MB-derived data (Bousigues et al., 2025; Camomilla et al., 2017), particularly during highly dynamic movements where errors could potentially increase (Dobos et al., 2022; Lahkar et al., 2022; Lapinski et al., 2019). Nonetheless, in the absence of a true gold standard such as bi-planar fluoroscopy, MB motion capture remains the most practical standard for assessing ML system accuracy (Wade et al., 2022). Finally, pose estimation accuracy depends heavily on the quality of keypoint detection, which may be influenced by factors such as clothing, lighting, and differences between test participants and the machine-vision algorithm's training data (Antognini et al., 2025; Kanko, Laende, Davis, et al.,

2021; Wade et al., 2022). Greater camera-to-volume distance could exacerbate these challenges; while Hawk-Eye deploys high-resolution, zoomed cameras (Sony Corporation, 2024), it remains unclear to what extent these settings mitigate such effects. Each ML system was evaluated in its standard operational configuration, so the observed differences in performance likely reflect combined effects of hardware arrangement, camera count, and pose estimation algorithms rather than any single factor (Scataglini et al., 2024; Wade et al., 2022; Yang et al., 2025). Further analyses could systematically manipulate camera configuration or standardise image data across ML systems to isolate their individual effects on pose estimation accuracy in sport-specific tasks. Despite their differing configurations, both ML systems produced comparable estimates for several kinematic variables. Hawk-Eye's widespread adoption across professional sports demonstrates its current ecological validity (Fitzpatrick et al., 2024; Michaelides, 2025), whereas Theia3D's consistent agreement with the MB reference across a broader range of kinematic and kinetic measures supports its versatility across in-lab and field-based settings.

Conclusion

This study evaluated the accuracy and reliability of an in-stadium and a portable multi-camera markerless motion capture system for quantifying baseball pitching biomechanics. The portable system demonstrated smaller errors in numerous measures, while the in-stadium system showed comparable agreement in several key variables and offers the practical advantage of in-competition data capture. These findings illustrate the trade-offs between criterion and ecological validity in markerless motion capture and emphasise the importance of continued validation of diverse system architectures across high-speed, sport-specific applications.

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Author contribution

A.L.A.: Conceptualisation, Writing – original draft, Writing – review & editing, Project administration, Methodology, Funding acquisition, Data curation, Formal analysis, Visualisation. **T.C.:** Writing – review & editing, Project administration, Methodology, Data curation, Visualisation. **D.T.L.:** Writing – review & editing, Methodology, Formal analysis. **R.F.E.:** Writing – review & editing,

Methodology. **J.H.B.:** Writing – review & editing, Methodology, Data curation.

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Declaration Generative AI

During the preparation of this manuscript, the authors used generative AI to check and improve the grammar of this work. The authors subsequently reviewed and edited the paper as needed but attest to its originality and take full responsibility of its content.

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